IA2 - AP_DS- Tutorial 04- Essential Python Libraries

Exercise 01: Temperature Analysis with NumPy

We have the following table of daily temperatures (°C) measured over **7 days** in **3 cities**:

Day	Alger	Annaba	Oran
1	22	25	20
2	24	27	21
3	19	20	22
4	25	29	28
5	26	30	27
6	21	21	23
7	20	26	25

1. Create a NumPy array `T` with the data above.
2. Compute and display:
   - the maximum temperature for each city,
   - the maximum temperature for each day,
   - the overall maximum temperature in the table.
3. Find the **day and city** where the overall maximum temperature occurred.
4. Create a new array showing, for each day, the **higher temperature between Alger and City Oran** (use `np.maximum`).
5. Add a **new row** representing the average temperature of each city over the 7 days.
6. Determine the **days where Annaba was the hottest city** (i.e., City Annaba's temperature exceeded both Alger and Oran).

Exercise 02: Student Grades Analysis with NumPy

You are given the grades of **100 students** in **5 modules**: `THL`, `DSS`, `DB`, `SE`, `RE`

Grades are integers between **0 and 20**. Your task is to perform statistical analysis and visualization using only **NumPy** and **Matplotlib**.
You will compute averages, variances, and standard deviations, apply boolean masks to filter grades, and visualize module comparisons using plots.

Data Initialization

1. Generate the grades of **100 students × 5 modules** using `np.random.randint`.
2. Fix the random seed with `np.random.seed(101)` to ensure reproducibility.
3. Store the grades in a NumPy array called `grades_of_students`.
4. Define the list of modules: `['THL', 'DSS', 'DB', 'SE', 'RE']`.

5. Display a preview of the grades (first 5 students).

Basic Statistics

1. Compute the **mean grade per student**.
2. Compute the **mean grade per module**.
3. Compute the **median grade per module**.
4. Compute the **variance** and **standard deviation** for each module.
5. Find the **overall minimum and maximum grade** in the dataset.
6. Compute the 25th, 50th, and 75th percentiles for the module THL.

Boolean Mask and Filtering

1. Create a **boolean mask** indicating grades between **10 and 15**.
2. Count how many grades fall within this range.
3. Extract all grades between 10 and 15 using the mask.

Visualization (Comparison Between Modules)

1. Use a **bar plot** to compare the **average grades per module**.
2. Plot a **box plot** (or multiple histograms) to show the distribution of grades per module.
3. Customize titles, labels, and colors for clarity.

Extra practice

1. Identify students who have **all grades above 15**.
2. Compute the **top 5 students** with the highest average grades.
3. Find the module with the **highest average grade**.

Exercice 03: Hospital Data (Alger, Annaba, Oran) using Pandas

1. Create a DataFrame with the following columns:

- `City` , `Department` , `Age` , `DaysAdmitted` , `DailyCost` , `Satisfaction` , `Readmitted`

City	Department	Age	DaysAdmitted	DailyCost	Satisfaction	Readmitted
Alger	Cardiology	45	5	200	4.5	0
Annaba	Neurology	60	8	300	3.8	1
Oran	Orthopedics	30	3	150	4.2	0
Alger	Cardiology	50	7	220	4.0	0
Oran	Neurology	40	4	280	4.1	1
Annaba	Orthopedics	70	6	160	3.5	1
Alger	Cardiology	35	2	210	4.7	0
Oran	Neurology	55	9	290	3.9	1

2. Save the initial DataFrame `df` to a CSV file named `hospital_original.csv` , preserving its index.

3. Replace the default numerical index with the letters `A` to `H`. Display the updated DataFrame.

4. Display all patients from index `C` to index `F`.

5. Display the patients at positions 1, 2 and 3.

6. Display the `City`, `Department` and `DailyCost` columns for patients labeled `B` through `F`, inclusive.

7. Create a new DataFrame containing patients older than 45 years whose `Satisfaction` is below 4. Display only their `City`, `Age`, `Satisfaction` and `Readmitted` values.

8. Create a new DataFrame named `df_neurology` containing only patients admitted to the `Neurology` department. Keep the columns `City`, `Age`, `DaysAdmitted` and `DailyCost`, and reset the index to start from 0. Then set `City` as the index and sort the rows by `DailyCost` in descending order.

9. Count the number of patients in each city using `df_neurology`. Display the resulting counts and create a bar chart to visualize the number of patients per city.

Exercice 04: Hospital Data (Alger, Annaba, Oran)

Using the existing hospital DataFrame df,

1. Load it and Display their columns
2. **Add columns**:
   - `TotalCost = DaysAdmitted * DailyCost`
   - `AgeGroup`:
     - `"Young"` if Age < 30
     - `"Middle"` if 30 ≤ Age ≤ 50
     - `"Senior"` if Age > 50
3. **Add, Rename, and Delete Columns**:
   - Add a column for a 10% **Bonus** based on the daily cost
   - Rename the column `DailyCost` → `DailyFee`
   - Delete the column `Bonus`
4. **Data Transformation**:
   - Create a column `HighCost` showing `"Yes"` if total cost exceeds 60,000, otherwise `"No"`
   - Create a column `SatisfactionLevel`:
     - `"Excellent"` if Satisfaction ≥ 8
     - `"Average"` if 5 ≤ Satisfaction < 8
     - `"Poor"` if Satisfaction < 5
5. **Ranking and Sorting**:
   - Add a column ranking patients by **TotalCost** (highest rank = 1)
   - Sort the dataset by **RiskScore** in descending order
6. **City Summary**:
   For each city, calculate:

- Average total cost
- Average satisfaction
- Readmission rate (percentage of patients readmitted)

7. **Risk Score Calculation**:
   Use the formula:
   [RiskScore = (DaysAdmitted \times 0.4) + (Age \times 0.3) + (100 - Satisfaction \times 10) \times 0.3] Then classify into a new column `RiskCategory`:

   - `"High"` if RiskScore > 60
   - `"Medium"` if 40 ≤ RiskScore ≤ 60
   - `"Low"` otherwise
8. **Find** the city with the **highest average RiskScore**.

9. **Apply Data Cleaning and Mapping**

   - Replace missing or invalid `Satisfaction` values with the mean of the column
   - Map city names (`Alger` , `Annaba` , `Oran`) to region codes (e.g., `DZ-16` , `DZ-23` , `DZ-31`)
   - Replace binary values in `Readmitted` column: `0 → "No"` , `1 → "Yes"`
10. **Pivot Analysis**

- Create a pivot table showing average `TotalCost` and `Satisfaction` for each city and department
- Compare the result with a grouped summary to discuss the difference between both methods

11. **Export Results**

- Export the main dataset to `hospital_risk.csv`
- Export the city summary to `city_summary.csv`